

Assessment of carbon-free fuel ammonia combustion with low methanol blends in reducing GHG emissions including N₂O

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ABSTRACT

Ammonia is presenting an ideal pathway towards reducing greenhouse gas (GHG) emissions by substituting traditional fuels to achieve carbon neutrality. However, the generation of N₂O during ammonia combustion weakens ammonia's carbon-free nature due to its GHG effect being 298 times greater than that of CO₂. To enhance the combustion efficiency of ammonia and preserve its carbon-free characteristics, the N₂O generation mechanism during ammonia combustion with low methanol blending ratios was investigated. A hybrid Perfectly Stirred Reactor-Plug Flow Reactor chemical reactor network was utilized, and the results indicate that N₂O emissions are much higher under lean ammonia conditions (especially for $\Phi = 0.7$ – 0.8) than under ammonia rich conditions ($\Phi = 1.1$ – 1.2). Moreover, the equivalence ratio region achieving low emissions of NO, unburned NH₃, and N₂O simultaneously is found at $\Phi = 1.1$ – 1.2 . This region enables a reduction in GHG emissions by over 70% compared to pure methanol combustion. The production and consumption rates of N₂O are controlled by the reaction $\text{NH} + \text{NO} = \text{N}_2\text{O} + \text{H}$. At low equivalence ratios, the mole fraction of H decreases, causing a separation between production and consumption rate curves and resulting in significant emissions of N₂O. Compared to pure ammonia combustion, the N₂O emissions are reduced by low methanol blends of 2%–10% through different mechanisms during lean and rich combustion conditions, which are controlled by H and NH, respectively. Under lean conditions, the presence of methanol enhances the production of H, which promotes the decomposition of N₂O. Under rich conditions, methanol reduces the formation of NH, thereby suppressing the generation of N₂O. The obtained low GHG emission control region of ammonia/methanol combustion is expected to support the ammonia fuel application in the future.

1. Introduction

In recent years, the use of traditional fossil fuels has significant influence on the greenhouse effect, drawing considerable attention from industry to academia (Joung et al., 2020; Dong et al., 2021). To realize a carbon-free society, continuous efforts have been made to explore alternative energy sources, such as solar photovoltaics and wind power. However, the requirements for high power, extended endurance, and reliable functionality in maritime vessels have firmly established the indispensable role of traditional internal combustion engines. Exploring efficient combustion of novel carbon-free fuels is a simple and practical approach to achieving decarbonization within the maritime industry.

Ammonia, recognized as a carbon-free fuel, holds great potential as an alternative for achieving mid- and long-term carbon reduction goals in marine engines (Kurien and Mittal, 2022; Boero et al., 2023). Projections by the IEA indicate that ammonia is expected to contribute up to 46% of the total energy consumption in the maritime industry by 2050 in a carbon-free society (Bouckaert et al., 2021).

Ammonia fuel has a higher energy density in comparison with other zero-carbon fuels, such as hydrogen (as illustrated in Table 1) (Koike et al., 2012). The energy content of liquid ammonia per unit volume is approximately 1.5 times that of liquid hydrogen. Additionally, ammonia's storage requirements are more flexible, as cooling it to 240 K under ambient pressure is sufficient for liquefaction, whereas hydrogen

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Table 1
Energy densities of ammonia and hydrogen.

	Liquid ammonia 0.1 MPa 240 K	Liquid ammonia 1.0 MPa 293 K	Liquid hydrogen 0.1 MPa 20 K	Compressed hydrogen 20 MPa 298 K
LHV (MJ/L)	12.7	11.21	8.5	1.9
LHV (MJ/kg)	18.6	18.6	120	120

requires cooling to 20 K. These relaxed storage conditions provide ammonia with a cost advantage for both short- and long-distance transportation. Furthermore, ammonia production is plentiful and holds potential for further expansion to meet the increasing global demand, although 85% of current production processes still result in non-zero CO₂ emissions (Zhang et al., 2020; Frigo and Gentili, 2013). Ammonia also exhibits high scalability and adaptability, making it compatible with existing fuel infrastructure and reducing the need for new supporting facilities.

It should also be noted that ammonia fuel encounters some challenges in practical applications. Early studies once cited it as “rough proof that ammonia does not support combustion or burn” (Buckley and Husa, 1962). The high ignition temperature, narrow flammability limits, and low laminar burning velocity (LBV) of ammonia restrict its combustion performance (Meng et al., 2023a, 2023b; Cui et al., 2022). One straightforward and effective approach to enhance combustion performance is through the utilization of dual-fuel technology, which involves combining ammonia with high-reactivity fuels. Numerous scholars have extensively studied the blended combustion of ammonia with various fuels, including diesel (Liu et al., 2022; Zhang et al., 2023a, 2023b), gasoline (Dong et al., 2024a), hydrogen (Wang et al., 2021a), ethanol (Li et al., 2023a), methane (Zhang et al., 2023c), and dimethyl ether (Meng et al., 2022). Among blended fuels, methanol, as a low-carbon and sustainable alcohol fuel, has significant advantages in terms of emissions reduction, which has gained favor (Ariemma et al., 2021; Vancouillie et al., 2013; Wang et al., 2023a; Meng et al., 2024).

In recent years, several scholars have conducted fundamental experiments sequentially (Zheng et al., 2022; Li et al., 2023b; Zhuang et al., 2024). Wang et al. (2021b) first conducted experiments to measure the LBV of ammonia-methanol premixed fuel using heat flux measurements. The range of experiments covered equivalence ratios from 0.7 to 1.8 and methanol blending ratios from 0% to 100%, at room temperature and medium temperatures. Li et al. (2022) provided the ignition delay times (IDTs) of ammonia-methanol premixed fuel under various conditions, including pressures of 20 bar and 40 bar, equivalence ratios ranging from 0.5 to 2.0, and temperatures ranging from 845K to 1100K. In the realm of internal combustion engine simulation and experimental studies, Wang et al. (2023b) numerically studied the combustion and emission performance of ammonia/methanol-fueled engines. The results of their research indicated that an optimal mixture ratio of 8:2 ammonia to methanol offers favorable emission characteristics. Based on visualization techniques, Wei et al. (2023) explored the ignition of ammonia-methanol premixed gas by a methanol pilot flame. The research suggested that blending methanol at approximately 30% energy fraction can ensure stable ignition of ammonia. Dong et al., 2023, 2024b's pre-chamber study achieved low NO_x emissions and high ammonia combustion rate under the condition of an orifice diameter of 3 mm and low-rich ammonia ($\phi = 1.1$). Zhang et al. (2024) investigated the combustion performance of ammonia in optical engines and demonstrated the beneficial effects of hydrogen and methanol enrichment on improving emissions of NH₃ and N₂O. In addition, several scholars have carried out chemical kinetics modeling on the combustion of ammonia/methanol (Li et al., 2022, 2023b; Zhuang et al., 2024; Wang et al., 2021b, 2023b; Giri et al., 2023; Xu et al., 2022). Nadiri et al. (2023) utilized the Reaction Mechanism

Generator (RMG) to propose four cross-reactions for ammonia/methanol combustion. In the author's previous work (Lu et al., 2023), a combustion reaction mechanism for ammonia/methanol was developed using these four cross-reactions, comprising 60 species and 399 reactions. This mechanism accurately predicted the LBVs and IDTs over a wide range of methanol blending ratios.

Despite the excitement around ammonia as a fuel with zero-carbon emissions and high combustion efficiency, its potential for greenhouse gas (GHG) emissions should not be overlooked (Meng et al., 2023c). Among the by-products of ammonia combustion, N₂O is a powerful GHG despite its low proportion. Studies have shown that the global warming potential (GWP) of N₂O is 298 times higher than that of CO₂ (Adewale et al., 2019). The direct or indirect conversion of only 0.4% of ammonia into N₂O is sufficient to offset the GHG reduction benefits of completely replacing fossil fuels (Alnasif et al., 2022). Currently, there is a growing academic focus on the issue of N₂O emissions caused by combustion (De las Obras-Loscertales et al., 2015; Liu et al., 2019; Xi et al., 2022; Hu et al., 2023). Colorado et al. (2017) conducted experimental research exploring the mechanisms of N₂O generation that do not involve fuel bound nitrogen, and confirmed that significant N₂O emissions occur under extremely lean combustion conditions. More scholars are focusing on the decomposition of N₂O in exhaust gases (Liu et al., 2020). Gonzalez et al. (2017) studied the N₂O decomposition over a 95% Pt/5% Rh knitted catalyst during ammonia combustion and found that the presence of N₂O is highly stable under industrial combustion temperatures. Okafor et al. (Okafor, 2023) found that at a lower temperature of 1300 K, N₂O is mainly consumed rapidly by reacting with NH₂ and NO₂. Yun et al. (2022) conducted experiments to simulate the decomposition of N₂O in the exhaust environment. They found that N₂O starts to rapidly decompose when the temperature rises to 1273 K. Unlike traditional fuels, the combustion of ammonia can lead to significant formation of N₂O, which involves fuel-bound nitrogen. However, there are still gaps in controlling N₂O emissions during the process of ammonia combustion. Hayakawa et al. (2023) studied the mechanism of N₂O formation in ammonia-hydrogen premixed flames and observed a rapid increase in N₂O mole fraction when the equivalence ratio was below 0.6. Although Wang et al. (2023b) reported that increasing the methanol blending ratio can control high N₂O emissions, they did not conduct any reaction kinetic analysis.

Given the important role of N₂O in the GHG emissions calculation of ammonia combustion, assessing the potential of carbon-free ammonia fuel in reducing GHG emissions, including N₂O, is crucial. There is a lack of systematic research on the production mechanism of N₂O during the combustion of ammonia-methanol blends. Furthermore, the specific effects of variables such as methanol blending ratio and temperature on N₂O emissions remain unclear, necessitating further investigation to reduce the N₂O emissions through different combustion strategies. With this background, this study aimed to investigate the emission characteristics of N₂O during ammonia combustion with low methanol blending ratios. Additionally, an assessment of the potential for reducing GHG emissions from ammonia fuel was conducted by converting N₂O emissions into equivalent CO₂ emissions. The influence of equivalence ratio and methanol blending ratio on N₂O formation was also analyzed. Chemical reaction kinetics analysis was conducted based on a hybrid Perfectly Stirred Reactor–Plug Flow Reactor (PSR–PFR) chemical reactor network (CRN). Rate of production (ROP) and sensitivity analysis were performed to understand the mechanisms of N₂O generation and reduction, supporting the application of ammonia as a fuel.

2. Computational details

2.1. Development of chemical reaction model and validation

This study utilized a previously established ammonia-methanol combustion reaction mechanism (Lu et al., 2023), derived from Okafor's ammonia-methane mechanism (Okafor et al., 2018). The derived

mechanism demonstrated accurate prediction of NO emission, LBVs, and IDTs across a broad range, as shown in Figs. S3–S5. However, there were still limitations in predicting N₂O emissions with this mechanism. Osipova et al. (2022) evaluated eight different mechanisms and found that appropriately modifying the reaction parameters for NO conversion to N₂O does not significantly impact the agreement between experimental data and numerical predictions of NO concentration. Therefore, the pre-exponential factor of the rate constant for $\text{NH} + \text{NO} \rightarrow \text{N}_2\text{O} + \text{H}$ was reduced by 31% to provide better agreement between experimental and numerical results for N₂O. In addition, the reaction $\text{N}_2\text{O} + \text{H} \rightarrow \text{N}_2 + \text{OH}$ from Mei's mechanism (Mei et al., 2021), which has shown good prediction for N₂O, was included to further enhance the calculation accu-

during the combustion process. The residence time of the flame is set to 5 ms in the flame zone based on the literature (Vijrumbana et al., 2023b). Moreover, the recirculation of product gases within the combustion zones of each stage is set at 20%, which is a widely adopted setting in similar studies (Valera-Medina et al., 2017; Rutar and Malte, 2002; Vijrumbana et al., 2023b).

The NH₃/CH₃OH inlet has a fixed mass flow rate of 0.8 g/s. The study investigates the impact of the methanol blending ratio on the emission characteristics of N₂O by changing the ratio of ammonia and methanol. The methanol blending ratio $x_{\text{CH}_3\text{OH}}$ (molar basis) in the premixed gas can be calculated using the global oxidation reaction equation, as shown in equation (1):



racy of N₂O. Table 2 presents the Arrhenius parameters of the rate constants for these reactions.

Using Ansys Chemkin Pro software, a 1-D premixed laminar flame model was utilized to simulate the flame structure numerically. Predictions of maximum N₂O concentration were obtained using the improved mechanism and other mechanisms (Okafor et al., 2018; Mei et al., 2021; Glarborg et al., 2018; Coppens et al., 2007; Zhang et al., 2021; Bertolino et al., 2021; Shrestha et al., 2021; Han et al., 2021) at pressures of 4 atm and 6 atm, as shown in Fig. 1. The experimental data was obtained from Osipova et al.'s (Osipova et al., 2022) research on the structure of ammonia-hydrogen flames. It is evident that the improved mechanism exhibits significant improvement in predicting N₂O compared to the Okafor mechanism, and the predictions based on the new mechanism were in good agreement with the experimental data. The details of the evolution in N₂O concentration are shown in Fig. S1.

2.2. Chemical reactor network (CRN) model

In all numerical simulations, a hybrid PSR-PFR CRN (as shown in Fig. 2) was employed to calculate the characteristics of N₂O emissions, following the common practice for modeling generic swirl burners (Vijrumbana et al., 2023a; Valera-Medina et al., 2017; Rutar and Malte, 2002). The CRN model, developed based on the work of Mashruk (Alnasif et al., 2022), consists of gas inlets, PSR-clusters, and a PFR-cluster. Specifically, the gas inlets supply the ammonia-methanol mixture and air to the burner, respectively. The PSR-cluster simulated the mixing and combustion characteristics within the burner. It consists of a mixing zone, a flame zone, and a central recirculation zone (CRZ). And the PFR-cluster was utilized to simulate the post-flame region. This study focused on the emission characteristics of N₂O in the flame zone

The effect of equivalence ratio on N₂O emissions is explored by changing the mass flow rates of the air inlet $\dot{m}_{\text{air}}$ under different conditions, as shown in Supplementary material-Reactant mass flow rates. The equivalence ratio can be expressed as:

$$\phi = \frac{\frac{\dot{m}_{\text{CH}_3\text{OH}+\text{NH}_3}}{\dot{m}_{\text{air}}}}{\left(\frac{\dot{m}_{\text{CH}_3\text{OH}+\text{NH}_3}}{\dot{m}_{\text{air}}}\right)_{st}} \quad (2)$$

in the formula, $\dot{m}_{\text{CH}_3\text{OH}+\text{NH}_3} = 0.8$ g/s. The initial temperature and pressure are 298 K and 1 atm, respectively, as listed in Table 3.

3. Results and discussion

3.1. GHG emission calculation of ammonia/methanol combustion

Fig. 3 illustrates the temperature variation in the flame zone at different methanol blending ratios. It can be observed that the temperature peaks when the equivalence ratio is 1.0. With minor methanol blends, the temperature did not show a significant increase, particularly under lean conditions. N₂O production begins at temperatures above 1250 K under lean combustion conditions of pure ammonia, reaching its peak around 1350 K (Zhou et al., 2023). The investigated temperature exceeded this range and was within the fluctuation zone where both the generation and consumption of N₂O occur with equal intensity. Therefore, equivalence ratio control and fuel injection have significant impacts in this range, requiring further analysis of reaction kinetics.

Fig. 4 illustrates the variation in major emissions of NO, unburned NH₃, N₂O, and CO₂ with different methanol blending ratios in the equivalence ratio range of 0.7–1.4. The variation of these four emissions with temperature did not show strong correlation. In Fig. 4(a), the highest level of NO was observed at the equivalence ratio of approximately 0.85, which agrees with Xu's observation (Xu et al., 2022). After that, NO emissions decreased as the equivalence ratio increased and remained at a relatively low level (<2000 ppm) after the equivalence ratio exceeded 1.1. In contrast, unburned NH₃ remained at a relatively low level (<1200 ppm) before the equivalence ratio of 1.0, and then increased rapidly after the equivalence ratio of 1.1. This is closely related to the equivalence ratio. After reaching the equivalence ratio of 1.0, ammonia was not completely oxidized by the air and remained abundant. The excess ammonia reacted with NO through denitrification, resulting in a rapid decrease in NO emissions. By adding a small amount of methanol, the emission of unburned NH₃ significantly improved. Adding just 10% (by molar fraction) of methanol was sufficient to reduce NH₃ emissions by 75% at $\phi = 1.1$. However, there was a slight deterioration in NO emissions, with the peak value of NO emission

Table 2

List of the reactions added or modified to improve N₂O predictions.

No.	Reaction	Units are cm, mol, s, cal in $k = AT^{\beta} \exp\left(\frac{-E}{RT}\right)$			Ref.
		A	β	E	
1	N ₂ O + H = N ₂ + OH	2.53E+10	0.00	4.55E+3	Mei et al. (2021)
	DUP	5.00E+14	0.00	1.81E+4	Mei et al. (2021)
2	NH + NO = N ₂ O + H	2.00E+14	−0.40	0.00	
	DUP	−2.20E+13	−0.23	0.00	
Original	NH + NO = N ₂ O + H	2.90E+14	−0.40	0.00	
Original	DUP	−2.20E+13	−0.23	0.00	

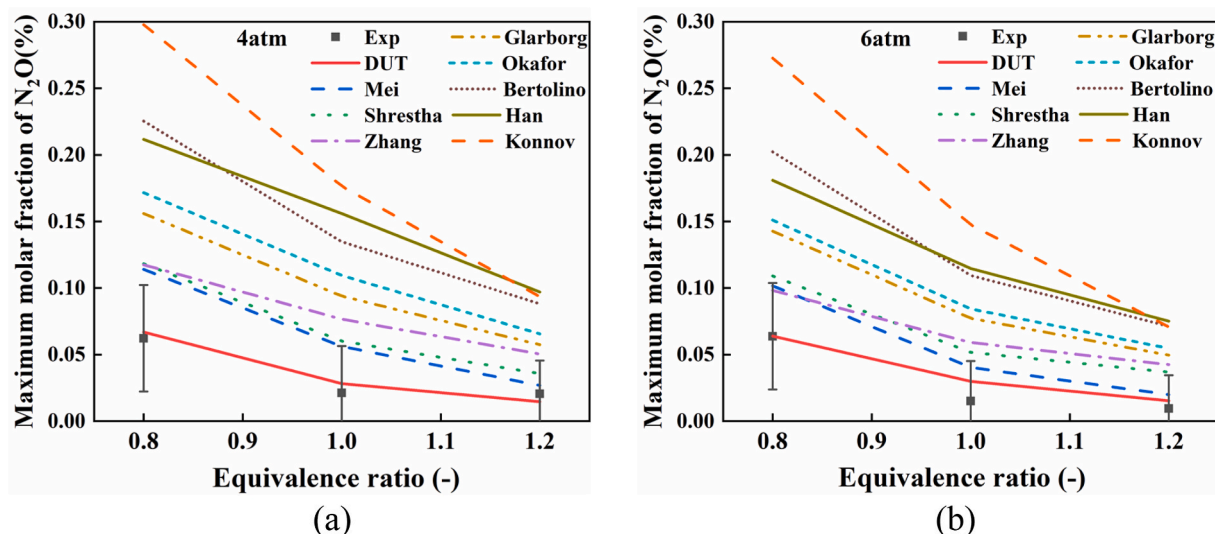


Fig. 1. Experimental and numerical values of peak concentration of N_2O at (a) 4 atm and (b) 6 atm.

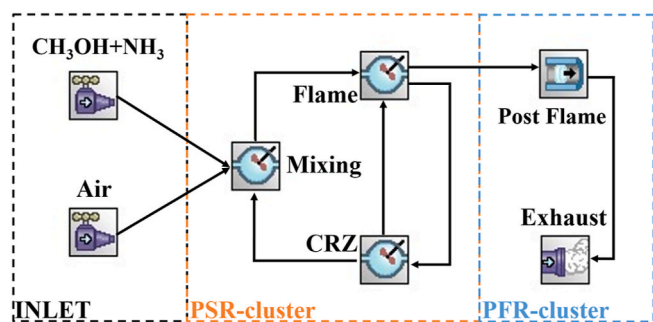


Fig. 2. PSR-PFR chemical reactor network used in the present study.

Table 3

Numerical simulation conditions used in the present work.

Variables	The simulation conditions
Equivalence ratio (ϕ)	0.7–1.4
Methanol blending ratio (x_{CH_3OH})	0%, 2%, 5%, 10%
Initial temperature (T_0)	298K
Initial pressure (P_0)	1 atm

increasing by 2500 ppm. Nevertheless, we could still achieve the desired low emissions of NH_3 and NO in the equivalence ratio range of 1.1–1.2 (as shown by the yellow region in Fig. 4(a)). With the addition of 10% (molar fraction) of methanol, NH_3 emissions decreased significantly by over 7500 ppm, reaching a level of 4500 ppm at $\phi = 1.2$, while the NO emission increased by only 340 ppm. It should be noted that this study focuses on purification during combustion rather than exhaust emissions, thus there is still potential for further reduction in these emissions.

In Fig. 4(b), the emission of N_2O was also related to the emissions of NH_3 and NO. Under lean conditions with sufficient oxygen, NH_3 was completely oxidized, and N_2O began to surge with NO, reaching its peak under extremely lean conditions ($\phi = 0.7$), and its minimum value between the equivalence ratio of 1.1–1.2 (88.5 ppm at $\phi = 1.1$ and 82.5 ppm at $\phi = 1.2$ under pure ammonia conditions). In this equivalence ratio range, low emissions of NO, N_2O , and unburned NH_3 can be achieved simultaneously. However, with the addition of methanol, N_2O showed an opposite trend to NO, decreasing with the increase in methanol blending ratio. With 10% methanol blends, the maximum reduction in N_2O emissions was 676 ppm, and the greatest percentage

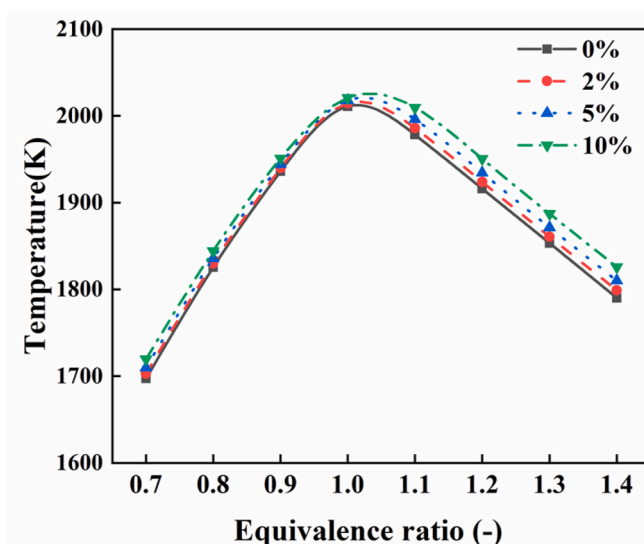


Fig. 3. Combustion temperature in the flame area with different methanol blending ratios.

decrease in N_2O emissions was 46.8% compared to the pure ammonia condition. This is an interesting phenomenon because N_2O is predominantly formed through the reaction $NH + NO \rightarrow N_2O + H$ from NO during ammonia combustion (Hayakawa et al., 2023). Further chemical reaction kinetics analysis is required to address this issue. CO_2 emissions were basically proportional to the methanol blending ratio and decreased slightly due to incomplete combustion at high equivalence ratios and oxidation to CO at low equivalence ratios.

N_2O and CO_2 are both significant contributors to greenhouse gas (GHG) emissions. During ammonia combustion, N_2O emissions, which involve fuel-bound nitrogen, offset the GHG reduction benefits of ammonia substitution for carbon-containing fuels. Hence, a combined assessment of N_2O and CO_2 emissions is required to evaluate the potential of NH_3 in reducing GHG emissions. The GWP of N_2O is 298 times greater than that of CO_2 (Adewale et al., 2019). Therefore, the GHG emission ($GHG_{equivalent}$) during ammonia-methanol co-combustion can be expressed as:

$$GHG_{equivalent} = 298 \times x_{N_2O} + x_{CO_2} \quad (3)$$

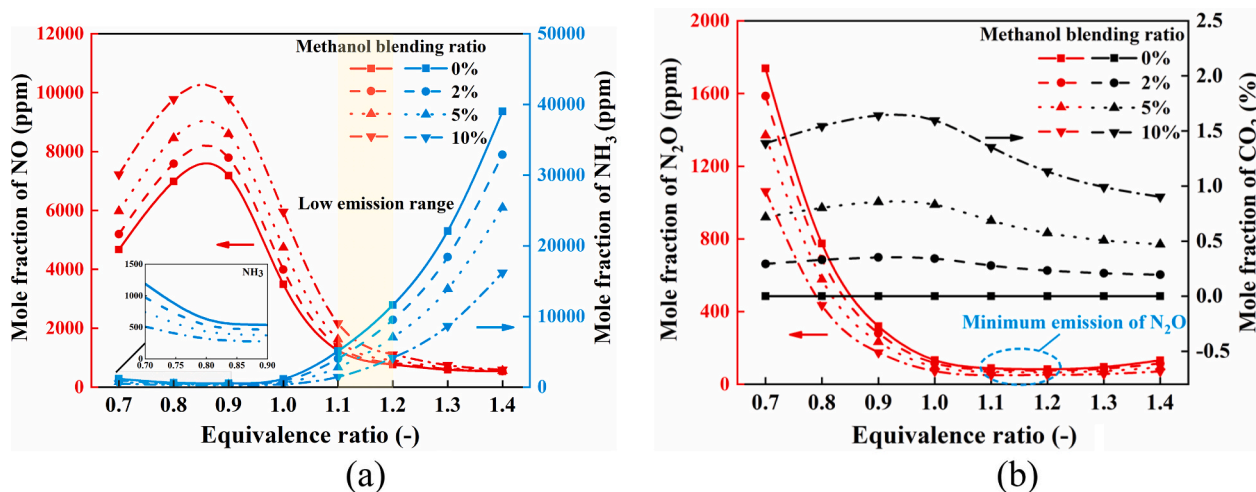


Fig. 4. Fraction of (a) NO, unburnt NH_3 , (b) N_2O and CO_2 in the flame area with different methanol blending ratios.

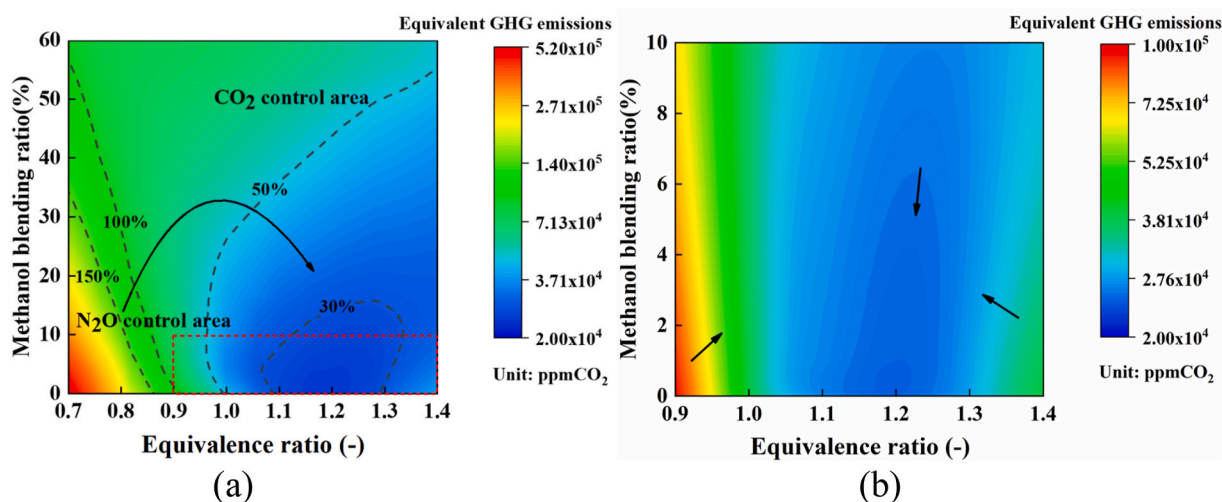


Fig. 5. Equivalent GHG emissions with different methanol blending ratios (The black arrow indicates the direction of GHG emission reduction).

in the formula, $x_{\text{N}_2\text{O}}$ is mole fraction of N_2O and x_{CO_2} is mole fraction of CO_2 .

Fig. 5 shows the emissions of equivalent GHG under different equivalence ratios with varying methanol blending ratios. The GHG emissions could be divided into three areas: the N_2O control area, CO_2 control area, and low GHG emission area. Under lean conditions, N_2O emissions dominated, resulting in the highest GHG emissions observed in this region. However, as the methanol blending ratio increased, GHG emissions gradually decreased. Under rich conditions, GHG emissions increased with increasing methanol blending ratio, and CO_2 emissions dominated. The transition between the two control areas occurs around the equivalence ratio of 0.95. To further assess the GHG reduction potential of ammonia, Fig. 5(a) used black dashed lines to represent the level of GHG emissions when burning pure methanol. The dashed line of 100% indicated that the GHG emission level was equivalent to that of pure methanol combustion at the same mass. On the right-hand side of the 100% dashed line, substituting methanol combustion with ammonia combustion had a positive impact on reducing GHG emissions, while on the left-hand side, it had a negative impact. As shown in Fig. 5(a), N_2O emissions cannot be avoided under lean conditions, and reduction of GHG emissions could only be ensured when the equivalence ratio was greater than 0.9 for all methanol blending ratios. However, by increasing the methanol blending ratio, the equivalence ratio range for

reducing GHG emissions could be extended. Even with a equivalence ratio of 0.7, when blending more than 60% of methanol, replacing methanol combustion with ammonia still positively contributed to controlling GHG emissions. The emissions of GHG decreased along the direction indicated by the arrow passing through the N_2O control area, CO_2 control area, and finally reaching the low GHG emission area (indicated by the red dashed line in Fig. 5(a)). Fig. 5(b) represents an enlarged view of this area, where GHG emissions decreased along the direction indicated by the arrow in the figure. GHG emissions reach their lowest value at $\phi = 1.2$ for pure ammonia combustion. Therefore, it can be concluded that blending a small amount of methanol under slightly rich conditions is beneficial for controlling GHG emissions.

3.2. N_2O generation mechanism of ammonia/methanol combustion

According to Fig. 5, significant variations in GHG emissions were observed along the equivalence ratio when the methanol blending ratio was 5%, ranging from unacceptable to ideal emissions. Therefore, a sensitivity analysis of N_2O emissions was conducted under different equivalence ratios when the methanol blending ratio was 5%, as shown in Fig. 6. The main reactions that impact N_2O emissions can be classified into three categories. The first category involved direct generation and consumption reactions of N_2O , such as the reactions of $\text{NH} + \text{NO} = \text{N}_2\text{O} + \text{H}$ and $\text{N}_2\text{O} + \text{H} = \text{N}_2 + \text{OH}$, which were the dominant reactions for

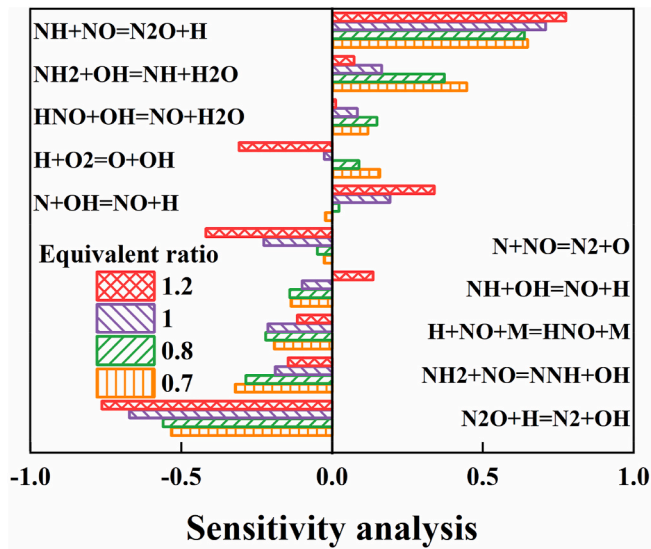


Fig. 6. Sensitivity analysis on N_2O at $x_{CH_3OH} = 5\%$ with different equivalence ratios.

N_2O generation and consumption. These reactions become less sensitive as the equivalence ratio increases. The second category encompasses reactions associated with fuel-NO formation or denitrification with amines, such as the reactions of $HNO + OH = NO + H_2O$, $N + NO = N_2 + O$, $H + NO + M = HNO + M$, and $NH_2 + NO = NNH + OH$. These reactions primarily govern the generation and consumption of NO, which

was the primary source of N_2O . The third category involved interconversion between OH and H, such as the reactions of $H + O_2 = O + OH$, $N + OH = NO + H$, and $NH + OH = NO + H$. Based on previous work (Lu et al., 2023), H would replace OH as the most important chain propagation carrier when the equivalence ratio exceeded 1.1 during ammonia-methanol premixed combustion. Therefore, as the equivalence ratio changes, the direction of these reactions also changed, which was reflected in the figure as sensitivity coefficients for N_2O emissions with different signs. Additionally, H was also the major reactive species involved in the generation and consumption reactions of N_2O . Its variation is likely to be the primary factor influencing N_2O emissions with changes in the equivalence ratio. Overall, the generation of H contributes to the consumption of N_2O .

Fig. 7 shows the reaction pathway of ammonia combustion at different equivalence ratios with a methanol blending ratio of 5%. The percentage values in the figure represented the relative rate of production (ROP) of each species, and the most significant active radicals involved in each reaction pathway were highlighted in red. Ammonia primarily reacted with OH to form NH_2 , which was subsequently consumed through three main pathways. The first pathway involved the coupling reaction of NH_2 to produce N_2H_2 , which became more significant in slightly reducing conditions (Glarborg et al., 2018). Clearly, as rich combustion conditions create a favorable high-temperature reducing environment, the first pathway steadily strengthens as the equivalence ratio increases, rising from 1.0% at $\phi = 1.0$ –7.3% at $\phi = 1.2$. The second pathway involved hydrogen abstraction reactions among amines, OH, and H active radicals. This pathway served as the primary consumption route for ammonia, and it gradually strengthened with the increase in equivalence ratio. Moreover, there was a change in

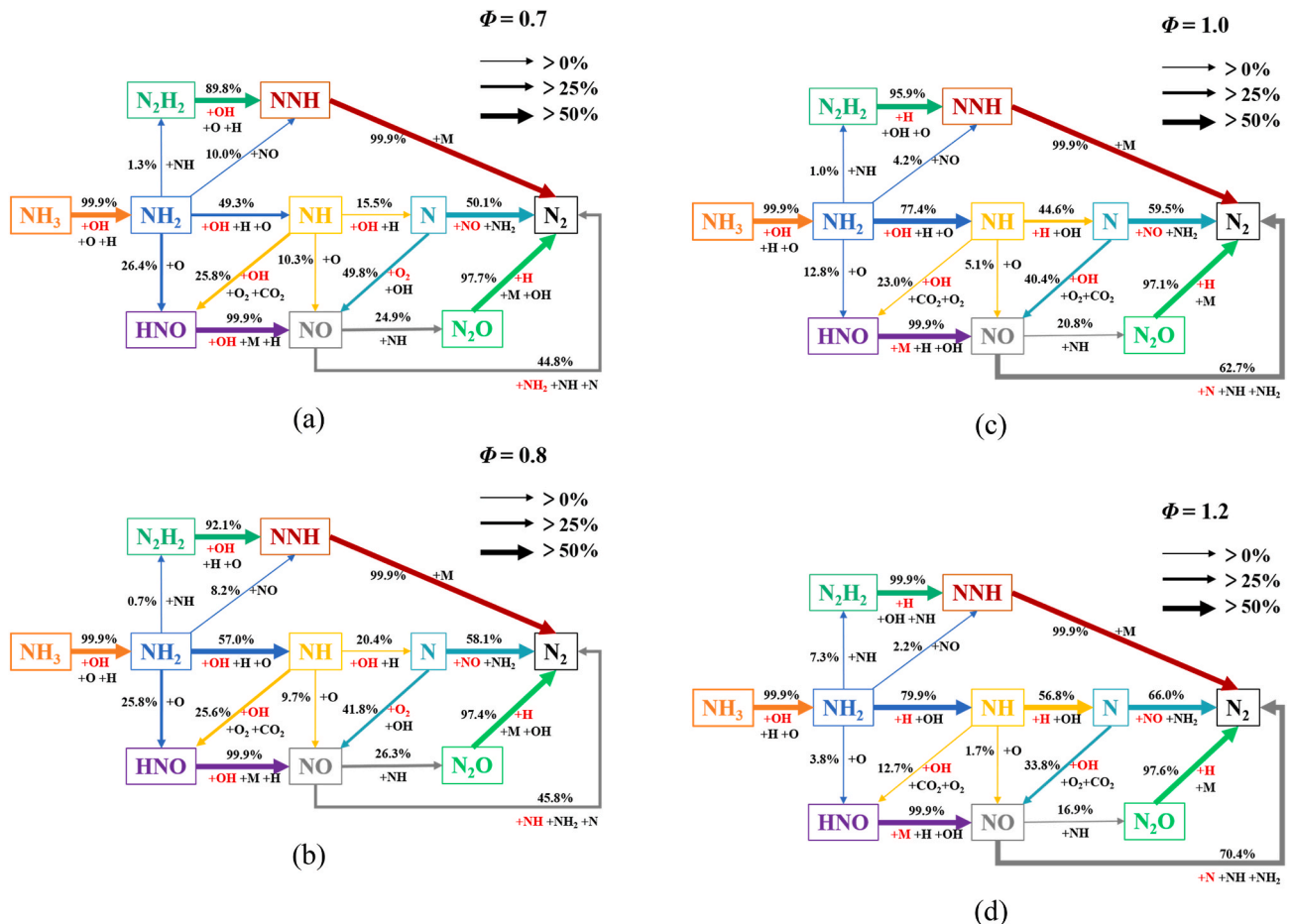


Fig. 7. Chemical reaction pathway at $x_{CH_3OH} = 5\%$ with different equivalence ratios.

the significant participating active radicals, transitioning from OH to H as analyzed in Fig. 6. The third pathway involved oxygen addition reactions between amines and O or O₂. This was the primary route for fuel NO generation, accounting for a larger proportion under oxygen-rich conditions and a smaller proportion under oxygen-deficient conditions. It was noteworthy that during the NO reduction reaction (denitrification reaction), the significant participating active radicals shifted from NH₂ to NH, and eventually to N, as the equivalence ratio increased. This could be attributed to the incomplete oxidation of ammonia, which consequently triggered the stepwise decomposition of excessive ammonia. As a result, the third pathway was weakened, leading to considerable consumption of ammonia through the second pathway. When focusing on N₂O, it was observed that its generation and consumption were primarily controlled by NH and H, respectively. With an increase in the equivalence ratio, the pathway for N₂O generation gradually weakened, while the consumption pathway was relatively stable. Therefore, it could be inferred that the emission of N₂O was primarily governed by the pathway in which NO converted to N₂O.

To further analyze the mechanism of N₂O generation and consumption, Fig. 8 examines the consumption, production, and net generation rates of N₂O with different methanol blending ratios. It was observed that the net generation and emission changes of N₂O exhibited consistency, remaining at a low level at high equivalence ratios but increasing sharply when the equivalence ratio was less than 1.0. The addition of a small amount of methanol delayed this upward trend. Decomposing the net generation rate of N₂O into its consumption and production rates revealed that both curves showed a variation pattern similar to that of NO, peaking at an equivalence ratio of around 0.85.

However, an equivalence ratio of 0.85 was not the point of maximum N₂O emission. It was observed from Fig. 8 that under rich combustion conditions, the consumption and production rates of N₂O remained nearly identical. This was because the consumption rate of N₂O was primarily controlled by the generation rate. In a highly reducing environment with enriched H, as shown in Fig. 9, N₂O was quickly consumed once it was formed. On the other hand, under lean combustion conditions, the concentration of H rapidly decreased, resulting in a separation between the consumption and production rate curves of N₂O, which led to significant N₂O emission. The significant participating radical in fuel NO formation, OH, reached its maximum value at around an equivalence ratio of 0.9 and indirectly controlled the consumption and production rates of N₂O.

3.3. N₂O generation control path of ammonia/methanol combustion

Fig. 10 shows that the addition of methanol facilitated the oxidation and decomposition of amines, resulting in more NH₂ converting into NH

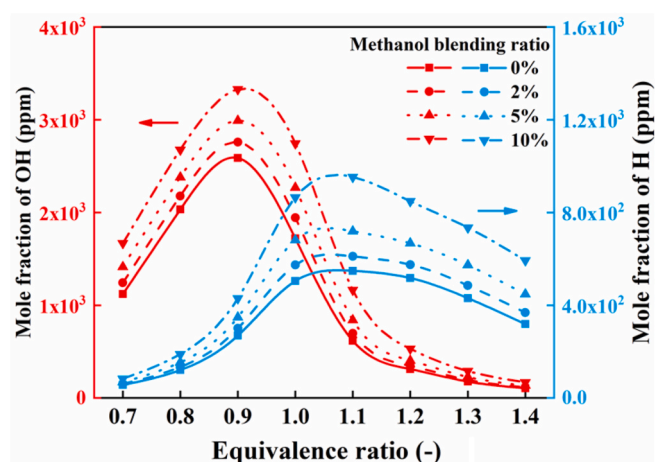


Fig. 9. Fraction of OH and H in the flame area with different methanol blending ratios.

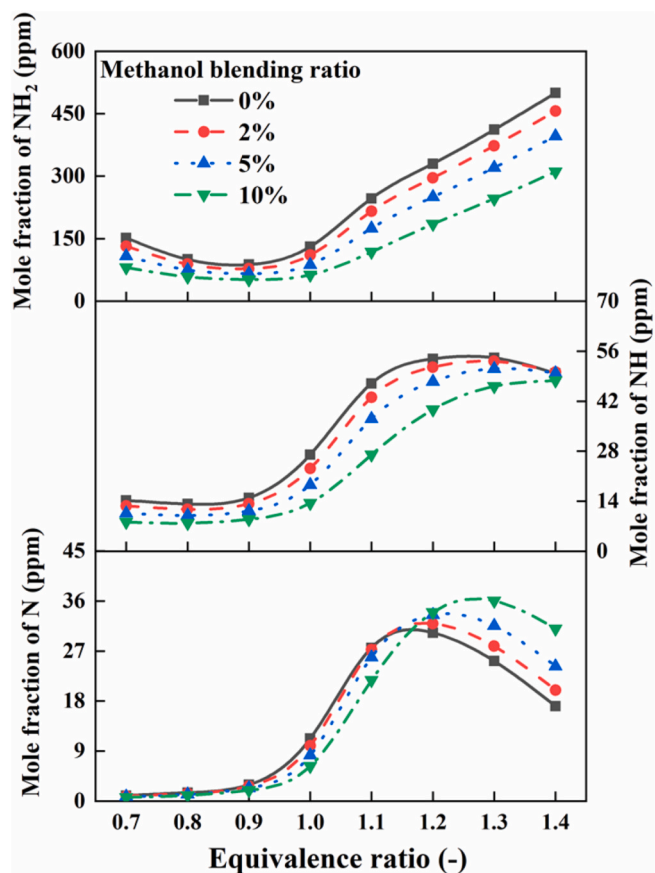


Fig. 10. Fraction of Primary, secondary, and tertiary amine in the flame area with different methanol blending ratios.

and N, particularly under fuel-rich conditions. However, with the addition of methanol, the concentration changes of amines did not remain consistent around $\phi = 1.1$, primarily due to two reasons. Since NH₂ was the primary decomposition product of NH₃, its concentration changed consistently with NH₃. Similarly, NH and N also decreased at low equivalence ratios due to methanol blends which enhanced combustion efficiency. On the other hand, the decomposition of amines required active radicals such as OH and H. At high equivalence ratios, these species were less abundant, which hindered amine decomposition. As a result, NH and N concentrations decreased as the equivalence ratio

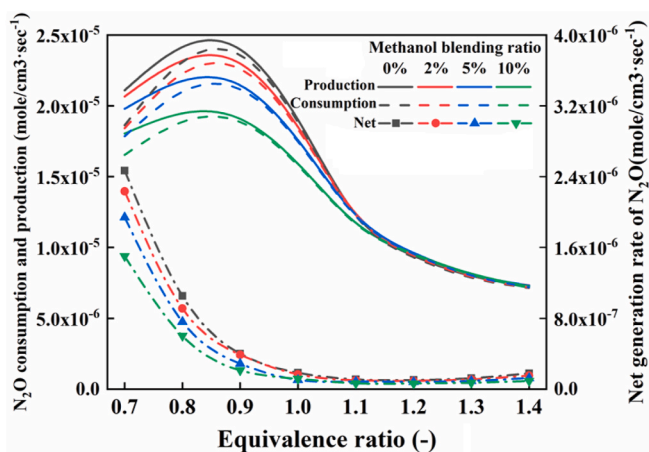


Fig. 8. N₂O consumption, production and net generation rate with different methanol blending ratios.

increases. Adding methanol increases OH and H levels, which promoted the decomposition of amines. Therefore, the combined effects of these factors resulted in the unique trend shown in Fig. 10.

Under oxygen-rich conditions, the addition of methanol led to the generation of more OH (as shown in Fig. 9), which enhanced the fuel-NO pathway and resulted in more conversion of amines to NO. The generation of N_2O was predominantly controlled by the reaction $NH + NO = N_2O + H$. Therefore, the addition of methanol in both conditions decreased the concentration of the main participating radical NH involved in N_2O formation, consequently reducing the production rate of N_2O (as shown in Fig. 8). Additionally, the addition of methanol increased the concentration of H, thereby narrowing the gap between the consumption and production rates of N_2O and ultimately reducing the net generation of N_2O . Furthermore, the rates of N_2O consumption and production were found to be largely unaffected by the addition of methanol when the equivalence ratio exceeded 1.1. This could be attributed to the opposite trends between NH (as shown in Fig. 10) and NO (as shown in Fig. 4), which offset the increase in N_2O consumption rate caused by the increase in NO fraction.

The generation of N_2O is primarily driven by the reaction $NH + NO = N_2O + H$, so further analysis is unnecessary. However, unique variations in N_2O consumption required ROP analysis, as shown in Fig. 11. Two main reactions were responsible for N_2O consumption: $N_2O + H = N_2 + OH$ and $N_2O (+M) = N_2 + O (+M)$.

The majority of N_2O consumption occurred through $N_2O + H = N_2 + OH$, with its reaction rate influenced by both N_2O and H. Furthermore, as the equivalence ratio and methanol blending ratio increased, the proportion of this reaction also increased. For example, at $x_{CH_3OH} = 5\%$, the proportion increased from 65% at $\phi = 0.7$ –82% at $\phi = 1.2$. Similarly, at $\phi = 0.8$, the proportion increased from 61% at $x_{CH_3OH} = 0\%$ –

68% at $x_{CH_3OH} = 10\%$. This suggested that the equivalence ratio had a comparatively greater impact on the reaction when compared to the methanol blending ratio. However, its influence was limited by the concentration of NO. At $\phi = 1.2$, although the reaction had the highest proportion, the reaction rate was the lowest. When the methanol blending ratio was 10% and the equivalence ratio was 1.2, the proportion reached a maximum of 84%, but the reaction rate was only $7.97 \times 10^{-6} \text{ mol/cm}^3 \cdot \text{sec}^{-1}$, which was lower than the reaction rate of $1.42 \times 10^{-5} \text{ mol/cm}^3 \cdot \text{sec}^{-1}$ at the minimum proportion of 61% ($x_{CH_3OH} = 0\%$ and $\phi = 0.8$).

The reaction $N_2O (+M) = N_2 + O (+M)$ accounted for the second-largest proportion of N_2O consumption, and both its reaction rate and proportion exhibited a trend similar to the N_2O production rate. At $x_{CH_3OH} = 0\%$ and $\phi = 0.8$, this reaction reached the maximum proportion of 33% and the maximum reaction rate of $7.57 \times 10^{-6} \text{ mol/cm}^3 \cdot \text{sec}^{-1}$. Conversely, at $x_{CH_3OH} = 10\%$ and $\phi = 1.2$, this reaction reached the minimum proportion of 13% and the minimum reaction rate of $1.27 \times 10^{-6} \text{ mol/cm}^3 \cdot \text{sec}^{-1}$.

To summarize, by adding active fuels such as methanol, the H content can be increased, or by increasing pressure and temperature, the particle concentration can be elevated, thereby promoting N_2O consumption to a certain extent. However, it should be noted that these methods may enhance combustion intensity and increase the concentration of OH, thereby contributing to the generation of NO. Therefore, when utilizing these methods, it is important to strike a balance between controlling N_2O emissions and NO generation.

Fig. 5(a) shows that under conditions of low methanol blending, the addition of methanol at $\phi = 0.8$ leads to a notable reduction in GHG emissions. Consequently, chemical reaction pathway analysis was also conducted at $\phi = 0.8$ with different methanol blending ratios as shown

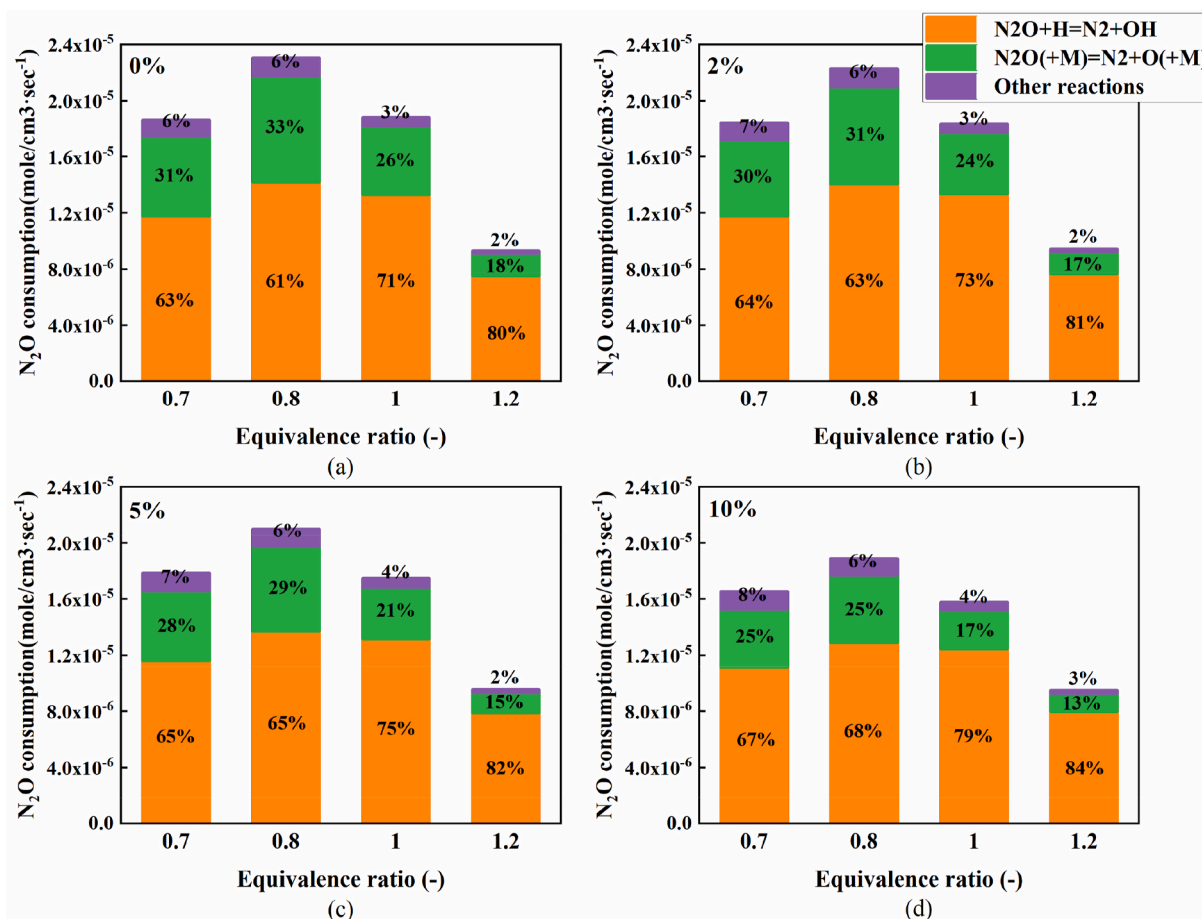


Fig. 11. N_2O ROP analysis with different methanol blending ratios.

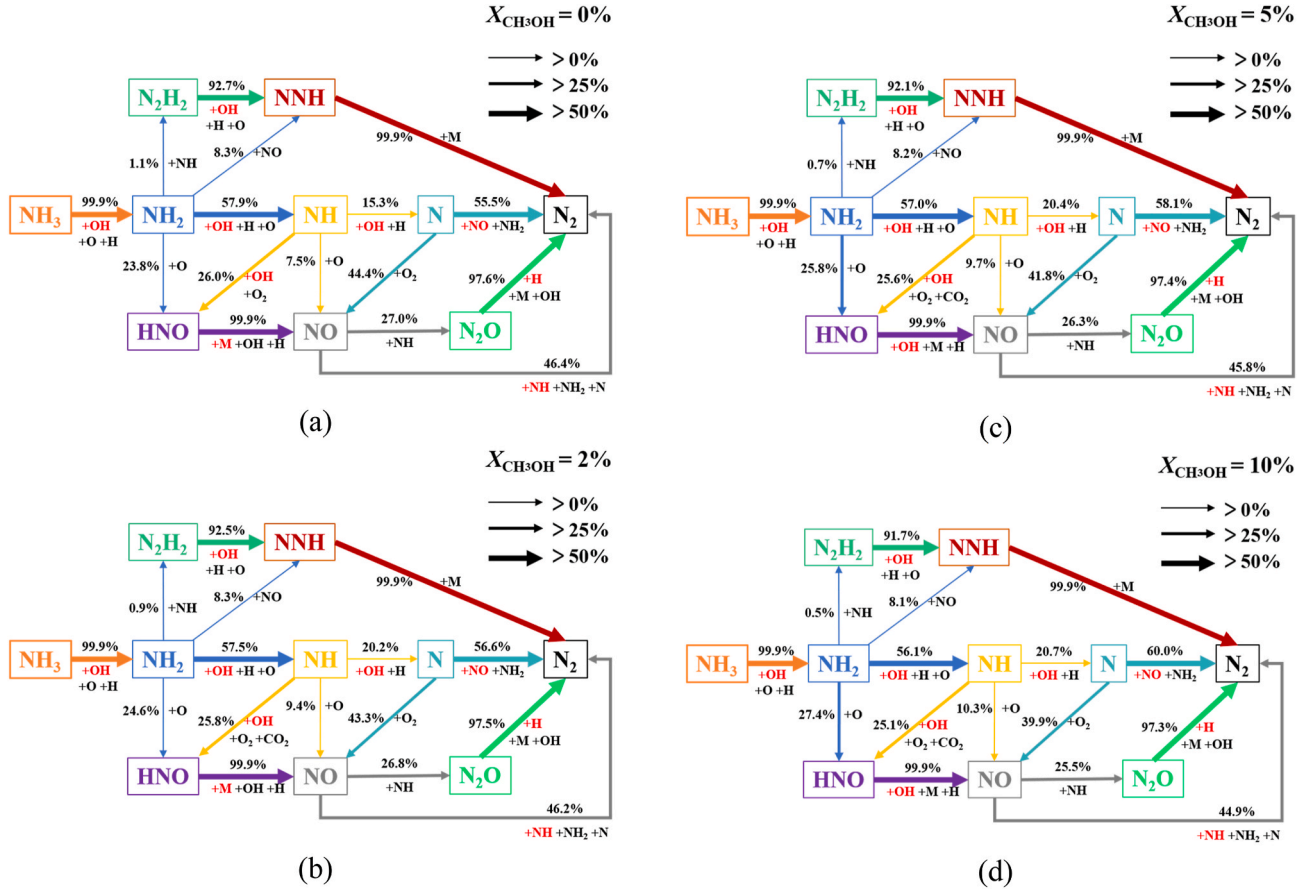


Fig. 12. Chemical reaction pathway at $\phi = 0.8$ with different methanol blending ratios.

in Fig. 12. It can be observed that with the increase in methanol blending ratio, the third pathway was enhanced, and the pathway of NH₂ to HNO rose from 23.8% at 0% methanol blending ratio to 27.4% at 10% methanol blending ratio. This is closely correlated with the intensified combustion activity resulting from the introduction of methanol. Furthermore, with the increase in methanol blending ratio, CO₂ was observed to participate as a reaction component in the pathway of NH to HNO. This observation provided additional insight into the reasons

behind methanol's promotion of fuel NO generation, despite the weakened second pathway leading to a declining trend in the strength of the pathway from NH to HNO. Furthermore, due to the weakening of pathway 2, the concentration of NH decreased, which led to a weakening of the pathway from NO to N₂O. Specifically, this decrease in the strength of the pathway was observed from 27.0% at 0% methanol blending ratio to 25.5% at 10% methanol blending ratio. With minimal changes in the pathway of N₂O consumption, the addition of methanol effectively controlled N₂O emissions. This finding complemented the separation of N₂O generation and consumption rate curves, as shown in Fig. 8.

Fig. 13 illustrates the sensitivity analysis of N₂O for different methanol blending ratios at $\phi = 0.8$. It can be observed that the addition of a small amount of methanol had a negligible impact on the overall reaction progress, as the sensitivities of different reactions exhibited insignificant variations. On this basis, the increased generation of H (as shown in Fig. 9) led to significant changes in reactions involving H, by enhancing the consumption reaction of H (such as N₂O + H = N₂ + OH, NO + H = NH + OH, and H + O₂ = O + OH) and weakening the production reaction of H (such as HNO + M = H + NO + M and N + OH = NO + H), resulting in the enhanced consumption rate of N₂O. In summary, the addition of methanol weakened the second pathway, thereby leading to a decrease in N₂O generation. In most cases, the reaction responsible for generating N₂O governed the process of N₂O consumption. However, the addition of methanol at low equivalence ratios (< 1.0) enhanced the N₂O consumption reaction, ultimately resulted in a reduction in N₂O emissions.

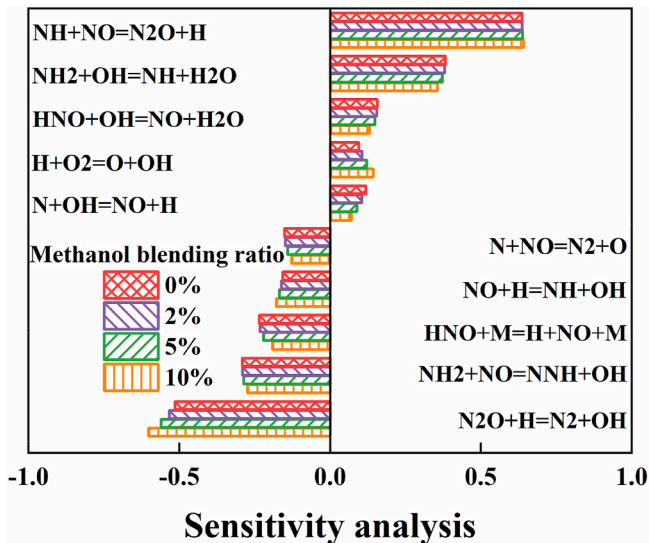


Fig. 13. Sensitivity analysis of N₂O at $\phi = 0.8$ with methanol blending ratios.

4. Conclusions

This study investigated the mechanism of N₂O emissions generation

during the ammonia combustion process with low methanol blending ratios to control GHG emissions and enhance combustion performance. The chemical mechanism was improved from the author's previous work (Dong et al., 2023), and the effects of equivalence ratio and methanol blending ratio were analyzed in a hybrid PSR-PFR chemical reactor network. The main conclusions are as follows:

- 1) Focusing on low methanol blending ratios, low-rich ammonia ($\Phi = 1.1$ – 1.2) reached minimum N_2O emissions between equivalence ratios of 1.1 and 1.2 (88.5 ppm at $\Phi = 1.1$ and 82.5 ppm at $\Phi = 1.2$ under pure NH_3 conditions), reducing greenhouse gas emissions by over 70% compared to pure methanol combustion. The impact of ammonia combustion on controlling greenhouse gas emissions is always positive when the equivalence ratio exceeds 0.9.
- 2) With the addition of 10% methanol, NH_3 emissions decreased significantly by over 7500 ppm at $\Phi = 1.2$, while NO emissions increased by only 340 ppm. The addition of methanol can also effectively reduce N_2O emissions. Under fuel-rich conditions, this is primarily achieved by increasing the fraction of the active radical H, which promotes the consumption of N_2O . However, under oxygen-rich conditions, the addition of methanol enhances the pathway for fuel NO generation, resulting in a decrease in NH concentration and suppression of N_2O formation.
- 3) The reaction rate of N_2O is predominantly controlled by the reaction of $\text{NH} + \text{NO} = \text{N}_2\text{O} + \text{H}$. Therefore, the trends in N_2O production and consumption rates are related to the variation of NO, which reaches the maximum around the equivalence ratio of 0.85. Under fuel-lean conditions, H is rapidly consumed through reactions (such as $\text{H} + \text{O}_2 = \text{O} + \text{OH}$), which leads to a separation between the consumption and production rate curves of N_2O , resulting in significant N_2O emissions.

CRedit authorship contribution statement

Mingfei Lu: Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Wuqiang Long:** Supervision, Resources, Project administration, Funding acquisition. **Fuxing Wei:** Writing – review & editing, Supervision, Methodology. **Dongsheng Dong:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Lixin Cong:** Supervision, Methodology. **Pengbo Dong:** Supervision, Funding acquisition. **Hua Tian:** Supervision, Project administration. **Xiaoyu Chen:** Data curation. **Shihao Chen:** Data curation. **Yongjian Wang:** Data curation. **Peng Wang:** Investigation, Data curation.

Declaration of competing interest

We declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2024.142755>.

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